Vehicle Control ECU - HLD v2

Title and Document Control

SYNTHETIC DATA FOR ACADEMIC PROJECT

Vehicle Control ECU - High Level Design

Document Version: v2

Revision: Introduces PowertrainCoordinatorSWC and renames EngineSpeedInterface to PowertrainSpeedInterface.

System Context

Vehicle Control ECU exchanges information with Engine Control ECU and Diagnostic Gateway. CAN is used for control traffic; Ethernet is used for diagnostic gateway uplink.

Software Components

VehicleControlSWC coordinates high-level vehicle behavior. EngineControlSWC manages engine-control commands. SensorManagerSWC aggregates sensor inputs. DiagnosticManagerSWC handles diagnostics. PowertrainCoordinatorSWC is a new coordination component.

Interfaces and Ports

PowertrainSpeedInterface carries EngineSpeed and EngineSpeedStatus. VehicleSpeedInterface carries VehicleSpeed. EngineControlSWC provides P_EngineSpeed and P_EngineStatus and requires R_VehicleSpeed. VehicleControlSWC requires R_EngineSpeed and provides P_VehicleCommand.

Dependencies

VehicleControlSWC now depends on PowertrainCoordinatorSWC for powertrain coordination and SensorManagerSWC for vehicle speed. DiagnosticManagerSWC reads health information from EngineControlSWC and SensorManagerSWC.

Revision Note

The previous revision used EngineSpeedInterface. This revision uses PowertrainSpeedInterface. Page 4 still refers to EngineSpeedInterface in a legacy note, creating a POTENTIAL INCONSISTENCY for review.

Communication

CAN channel CAN0 carries PowertrainSpeedInterface, VehicleSpeedInterface and VehicleCommand signals.